Lesson notes · for you

Choices, limits and a goal

Optimization · Level 2 · Lesson 1

Time	Materials
20 min	Paper and a pen; A price list from a shop or the school kiosk (or make one up)

Learning goals

- Write choices as variables with units.
- Write limits as inequalities and the goal as an expression.
- Check whether a plan fits by putting its numbers in.

Activity: The trip snack model

1. Choose (3 min): Pick three snacks for a school trip and write a price for each (real or made up), and how much money you have.
2. Write (7 min): Give each snack a letter, with its unit (packets, bottles). Write the money limit, one more limit such as “at least one drink”, and a goal such as “as many items as possible”.
3. Check (7 min): Write three plans. Put each plan’s numbers into every limit and mark it feasible or not. Which feasible plan scores best on your goal?
4. Share (3 min): Swap models with a partner. Find a rule their model is missing, such as sharing, a healthy choice or an allergy.

Use small, round prices so you can check plans in your head. No shop nearby? Make up the prices.

Discussion

- Why are prices data and not variables?
- When is the cheapest plan not the best one?
- Which rule did your partner’s model forget?

Check yourself

- One limit from my model, written as an inequality.

Try it this week
Write a model for a choice of your own, such as snacks for a school trip with your pocket money. Name the variables with their units, write the limits with ≥ or ≤, and write the goal. Then check two plans by putting in the numbers.

Worksheet

Choices, limits and a goal

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Name _____ Class _____ Date _____

1 **Each pupil gets at most 2 handfuls of groundnuts. Which do you write?**

◯ n ≥ 2 ◯ n = 700 ◯ n ≤ 2

2 **r = cups of rice, n = handfuls of groundnuts, g = cups of greens. Match each rule to its maths.**

At least 700 kcal of energy	• • • •	n ≤ 2
At most 2 handfuls of groundnuts	• • • •	260r + 170n + 40g ≥ 700
At least 5 mg of iron	• • • •	15r + 8n + 10g
The cost, as small as possible	• • • •	0.4r + 0.6n + 6g ≥ 5

3 **A plate with 2 cups of rice and 1 handful of groundnuts meets the energy limit of 700 kcal.**

◯ True ◯ False

4 **Which of these are limits, not choices or goals? Choose all that apply.**

☐ At most 2 portions of any one food ☐ Iron at least 5 mg ☐ Protein at least 20 g

☐ Cups of rice on the plate ☐ Make the cost as small as possible

5 **Complete the sentence.**

A _____ is a choice with a unit, a _____ is a rule written as an inequality, and the _____ is what you make as small or as large as possible.

Word bank: price · corner · limit · variable · goal

✎ **Think of a choice you make often: snacks, phone data, time for homework. What are its variables, limits and goal?**

Check your answers

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1 **$n \leq 2$**

"At most 2" allows 0, 1 or 2 handfuls, so the number of handfuls is less than or equal to 2: $n \leq 2$.

2 **At least 700 kcal of energy — $260r + 170n + 40g \geq 700$**
At most 2 handfuls of groundnuts — $n \leq 2$
At least 5 mg of iron — $0.4r + 0.6n + 6g \geq 5$
The cost, as small as possible — $15r + 8n + 10g$

Each "at least" limit adds what every food gives and uses $\geq$; "at most" uses $\leq$.

3 **False**

$260 \times 2 + 170 \times 1 = 690$ kcal, which is 10 short of 700.

4 **Iron at least 5 mg; At most 2 portions of any one food; Protein at least 20 g**

Limits are rules every plan must keep: iron, protein and the 2-portion rule.

5 **A variable is a choice with a unit, a limit is a rule written as an inequality, and the goal is what you make as small or as large as possible.**

These are the three parts of every model: variables (the choices), limits (the rules, as inequalities) and a goal (the one thing to make small or large).